FINAL EXAMINATION
IT31-DB2-DATABASE SYSTEM MANAGEMENT 2

# Guidelines and Requirements

## General Instructions:

Create and publish a functional web-based information system that stores and retrieves data through a backend connected to **MongoDB** or any **database management system** of their choice. The frontend must be deployed on **Vercel** so that the instructor can access and test the system online. Grading **focuses** on the database and backend, including database connection, data modeling and normalization, ERD, data integrity, working backend operations, the group presentation, and each member's individual knowledge and contribution.

## Requirements:

- Solve a clear record-management problem such as inventory, sales, appointments, library records, enrollment, service requests, or another approved topic.
- Use at least three meaningful and related collections. Each collection must have a clear purpose and suitable fields.
- Implement **Create, Read, Update, and Delete** operations for the main records. Operations must affect the MongoDB database, not temporary or hard-coded data.
- Use a **backend/API** as the controlled connection between the web interface and MongoDB. The browser must not contain exposed database credentials.
- Apply server-side validation, correct data types, required fields, duplicate prevention where appropriate, and clear error handling.
- Use **environment variables** for connection strings and secrets. Do not submit usernames, passwords, or secret keys in public source code.
- Prepare a logical **ERD** showing entities or collections, attributes, identifiers, and relationships with correct cardinality.
- Show **normalization** up to Third Normal Form as a logical design exercise, then explain how the normalized design was mapped to MongoDB using references or justified embedding.
- Provide sufficient sample data to demonstrate relationships, validation, queries, and all CRUD operations during evaluation.

**Normalization and Database Modeling**

Although **MongoDB** is a document database, normalization is still assessed as evidence of sound database design, and if the students used a **SQL** database it will be easier for them to create the database model. Students must first present a normalized logical model up to Third Normal Form. They must then explain their MongoDB or SQL database implementation choices.

**Required Submission**

- **Vercel** frontend URL that allows the instructor to test the system.
- Live **backend/API** URL and a short list of available endpoints or operations.
- **MongoDB** or any **database management system** database evidence: collection names, representative documents, validation rules or indexes, and working data changes.
- **ERD** in PDF or image format with readable entities or collections, fields, identifiers, relationships, and cardinalities.
- **Normalization evidence** showing the transformation from unnormalized data to 1NF, 2NF, and 3NF, followed by a brief MongoDB mapping explanation.
- **Source-code repository or compressed project folder**, including a safe sample environment file such as. env.example without real credentials.
- A short demonstration of the database connection, CRUD operations, validation, relationship handling, and deployed integration.

**Evaluation Rubric**

**Group/Application Grade**

| Criteria | Points | What is Expected |
|---|---|---|
| Database Connection and Security | 5 | Backend connects reliably to MongoDB or any DBMS in the deployed system. Connection settings use environment variables; credentials are not exposed. Connection failures are handled clearly. |
| Database Design and Normalization | 5 | Logical design reaches 3NF with correct dependencies and minimal redundancy. MongoDB collection structure is appropriate, and embedding or referencing decisions are clearly justified. |
| ERD Accuracy and Completeness | 6 | ERD is readable and matches the implemented database. It includes entities or collections, key attributes, identifiers, relationships, and correct cardinalities. |
| Backend CRUD and Data Operations | 6 | Create, read, update, and delete operations work correctly through the backend/API and persist in MongoDB. Related records and useful queries operate correctly. |

| Criteria | Points | What is Expected |
|---|---|---|
| Data Integrity Validation and Error Handling | 4 | Server-side validation, correct data types, required fields, duplicate controls or indexes, and useful error responses protect database quality. |
| Deployment Integration | 5 | The Vercel frontend and live backend work together. Submitted links are accessible and the deployed system reliably performs database operations. |
| Group Presentation and System Demonstration | 4 | The group gives an organized presentation and clearly demonstrates the connection, CRUD operations, validation, relationships, and deployed workflow. |

**Individual Presentation Grade**

| Criteria | Points | Guide | Score |
|---|---|---|---|
| Participation and role in presentation | 5 | Actively presents and handles an assigned portion of the application explanation. | |
| Clarity and organization of explanation | 4 | Explains ideas in a clear, logical, and organized manner. | |
| Knowledge of the application | 3 | Shows understanding of the features and functions being presented. | |
| Communication and confidence | 3 | Communicates clearly, professionally, and confidently. | |

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